

Exercises

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Surname, First name

Applied Microeconomics (FEB12001X)
 FEB12001X - Midterm 20 September 2023

1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
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General information

Lecturer: Josse Delfgaauw and Robert Dur

Date & Time of examination: September 20, 18:30 - 19:30

Type of examination: Closed book, Open questions

Number of questions: 3 questions (question 4 is additional answer space)

Number of pages: 12 pages (incl. cover sheet). Separate appendix: no.

Instructions

- Please use only a pen with black or blue ink.
- Fill in your name and student number as indicated above. **Also, code your student number by coloring the circles with the corresponding numbers.** If you made a mistake, put a cross through the wrong number, and color the correct one.
- Write your answer **in** the answer box corresponding to the question.
- **You have to provide a calculation or explanation of each answer.** An answer without calculation or explanation yields zero points.
- Because the total answer space is limited, we strongly advise you to prepare your answers on scrap paper: **no additional exam paper will be provided.**
- In case you made a mistake and therefore you need more space, use the additional writing space at the end of the exam. Clearly indicate where the (remaining part of the) answer can be found.
- Exam questions and answers, scrap paper and other exam related documents may not leave the exam room and may not be copied or disseminated in any way.
- It is not allowed to keep a watch within reach during the examination, it must be stored in your coat or bag.
- Any kind of mobile devices or (potential) data media must be switched off during the examination. All devices need to be stored in your coat or bag.

Permitted materials

- You are allowed to use a pen and blank scrap paper.
- You are allowed to use a calculator (only the following types: FX-82SX Plus / FX-82MS / FX-350MS / TI-30XA/B/S).
- You are NOT allowed to use notes.
- You are NOT allowed to use books.
- You are NOT allowed to use a dictionary.

Good luck!

Question 1

Suppose there are two individuals, Anne and Ben, who both derive utility from the consumption of a public good and from the consumption of a private good. Anne's marginal rate of substitution between the public good and the private good is given by

$$MRS_A = \frac{x_A}{G_A + G_B}$$

where x_A is the number of units of the private good that Anne consumes, G_A is the number of units of the public good bought by Anne, and G_B is the number of units of the public good bought by Ben. Ben's marginal rate of substitution is given by

$$MRS_B = \frac{x_B}{G_A + G_B}$$

where x_B is the number of units of the private good that Ben consumes. Marginal production cost of both goods are constant and equal to their price on the market. The price of the public good equals p_G . The price of the private good equals $p_x = 1$. As a result, the marginal rate of transformation between the public good and the private good equals p_G . Anne and Ben have the same level of income y , which they spend fully on the two goods. Given $p_x = 1$, this implies that $x_A = y - p_G G_A$ and $x_B = y - p_G G_B$.

- 1p **1a** Consider the assumptions of this exercise. How can you see in this exercise that the public good is non-rival while the private good is rival?

- 1p **1b** Determine the socially efficient level of units of the public good $G_A + G_B$.

Suppose that the public good is provided by the market. Anne and Ben decide independently how many units of the public good to buy (of course taking the presence of the other into account).

- 1p **1c** Determine the Nash-equilibrium that arises on the market (in terms of provision of the public good by Anne and by Ben).

Question 2

Consider two neighbours, Scott and Jacinda, who live in a country where people are allowed to build campfires in their back yard. If Scott builds a campfire in his back yard, smoke blows to Jacinda's back yard. Jacinda can choose to sit inside or outside (in her back yard). The table below gives the payoffs (in euro's) for each combination of decisions, where the first entry in each cell is Jacinda's payoff and the second entry Scott's payoff.

		Scott	
		Fire	No fire
Jacinda	Inside	10 , 25	10 , 0
	Outside	5 , 40	50 , 15

Scott and Jacinda maximise their own payoff.

- 1p **2a** Explain why the Nash-equilibrium differs from the socially efficient outcome. Use maximally four sentences and use the correct market failure in your explanation.



- 1p **2b** It is likely that both conditions of the Coase Theorem are met in the situation described in this exercise. Explain why in maximally four sentences.

Question 3

Company Z employs one worker. The production is described by the following function: $Q = E + K$, where $0 \leq K \leq 1$ and E is the effort chosen by the worker. The price of the product is equal to P . Note that K is an amount of production that is realized independent of the effort of the worker. We assume that $P > 2K$. The worker's utility function is $U = Y - E^2$ and the income Y of the worker is $Y = bQ$. Note that, by assumption, there is no base salary. When the worker chooses not to work for this company, his utility is equal to zero.

- 2p **3a** Derive the profit-maximizing bonus b . Do not only provide calculations, but also write a few sentences along the way explaining what you are doing.

- 1p **3b** Explain in a few clear sentences why the level of the profit-maximizing bonus decreases with K .

Assume from now on that, in addition to the bonus b , the company also pays a base salary, denoted by a . Hence, the employment contract now is: $Y = a + bQ$. The base salary can be negative.

- 1p **3c** Show that the base salary a that is needed to attract the worker is equal to $-Kb(b^2)/4$.

- 1p **3d** Explain in a few clear sentences why the addition of a base salary to the contract increases the profit-maximizing bonus b . You can derive the profit-maximizing bonus to discover the reason, but a derivation is not necessary for the answer to be correct.

Additional answer space

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found**. Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

4a Additional answer box 1

4b Additional answer box 2

4c Additional answer box 3

4d Additional answer box 4

4e Additional answer box 5

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